Nirbhay Shingala

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EDUCATION

Nirma University Expected 2028
Bachelor of Technology in Computer Science and Engineering CGPA: **9.04**

Relevant Courses: Data Structures and Algorithms, Computer Networking, Operating Systems, Object-Oriented Programming, DBMS, Software Development Life Cycle (SDLC), Computer Architecture

SKILLS

Technical Skills	Data Structures, Algorithms, DBMS, Machine Learning, Data Analysis, Data Visualization, EDA, Statistical Analysis, REST APIs, HTTP, WebSockets
Frameworks	FastAPI, Flask, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, React.js, Node.js, Express.js
Languages	Python , SQL , C++, JavaScript
Tools	Git, GitHub, Docker, Redis, gRPC, MongoDB, Vercel

EXPERIENCE

Computer Society of India — Executive Member **January 2025 - Present — Ahmedabad, India**

- **Executive Member, CSI Nirma University:** Coordinated **DIV WAR** and **four technical events at Cubix 2026**, managing event logistics and participation for **200+ students**.

PROJECTS

SuryaKiran – AI-Powered Solar Inverter Predictive Maintenance Platform GitHub Link

- Developed an AI-powered solar inverter analytics platform using **Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, and CatBoost** to analyze telemetry data and predict inverter faults.
- Engineered **25 time-series features** including lag features, rolling statistics, z-scores, power deltas, temperature trends, and physical efficiency ratios to capture early-stage inverter degradation.
- Built a **4-model weighted ensemble** combining Random Forest, XGBoost, LightGBM, and CatBoost for fault prediction, with **root-cause analysis, time-to-failure estimation, and real-time analytics dashboards** for actionable maintenance insights.

RelayMesh – Real-Time Messaging System GitHub Link | Demo Link

- Engineered a scalable real-time messaging platform using **React.js**, **Node.js**, **Express.js**, **MongoDB**, **Socket.IO**, and **Redis**, supporting one-to-one messaging, contact management, typing indicators, online presence, message history, delivery/read receipts, and offline message synchronization.
- Implemented distributed WebSocket communication using the **@socket.io/redis-adapter**, enabling horizontal scaling, synchronized presence updates, and reliable socket event propagation across multiple backend instances.
- Designed a secure backend with **JWT authentication**, **Google OAuth 2.0**, persistent chat storage, and integrated **Grafana** dashboards to monitor application health, system metrics, and real-time observability.

MailSense – AI-Powered Email Intelligence Platform GitHub Link

- Architected an AI-powered email intelligence platform using **React.js**, **Node.js**, **FastAPI**, **gRPC**, **MongoDB**, and **ChromaDB**, enabling semantic search, retrieval-augmented question answering, and explainable email prioritization.
- Implemented a hybrid retrieval pipeline combining **BM25**, dense vector search, **Reciprocal Rank Fusion (RRF)**, and **Cross-Encoder reranking**, supporting Gmail-style operators while achieving **p95 search latency below 500 ms across 100K+ indexed emails**.
- Integrated **Google Pub/Sub**, **Gmail API**, **Google Calendar API**, and **Twilio** to build an event-driven workflow that ingests incoming emails in real time, extracts actionable deadlines using AI, creates calendar events, and delivers automated SMS reminders.

LEADERSHIP & TEAMWORK

- **HackaMind Hackathon — Team Lead • Led the development of SuryaKiran** during a **48-hour hackathon**, building ML models using **Python and Flask**, securing a **top 10 in our track among 2300+ participants**.

ACHIEVEMENTS

- **LeetCode** • Solved **250+ DSA problems**, achieved a **max rating of 1567**, and ranked **5,296 out of 29,000+ participants** in **Weekly Contest 476**. (Profile Link)
- **Competitive Programming** • Solved **150+ DSA problems on Codeforces** and **400+ problems across LeetCode, Codeforces** (Profile Link).